

Upgrade Goldengate Hub from RHEL6 to RHEL7

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1. 1. Overview

This document is to provide a step-by-step on how to upgrade Goldengate Hub with the following changes:

- Upgrade Linux from RHEL6 to RHEL7
- Change Secrets CLI to IDPS

2. 2. Pre-Setup: Onboard IDPS (Once Per AWS Account. Completed for PSP)

- Follow <https://wiki.intuit.com/display/ISKM/IDPS+User%27s+Guide> to onboard IPDS. Need to onboard IPDS once per AWS account.

IDPS Onboard

1 The Basics

2 Application Details

3 Performance

4 Extras

Your Name

Simon Liu

Your Corp Username

sliu

Project Name

QBORDS

Asset ID

362557362305194983

Business Unit

SBG

Environment

PRODUCTION

BACK

NEXT

ATTENTION: Playground environments are now deprecated. All existing playgrounds have been deleted. Please use PRE-PRODUCTION for any non-prod environment

IDPS Onboard

1 The Basics

2 Application Details

3 Performance

4 Extras

Hosting Location

AWS

Current Encryption Strategy

Secrets CLI

BACK

NEXT

ATTENTION: Playground environments are now deprecated. All existing playgrounds have been deleted. Please use PRE-PRODUCTION for any non-prod environment

IDPS Onboard

✓ The Basics — ✓ Application Details — 3 Performance — 4 Extras

Max number of encryption keys

100

Max number of secrets

100

Max number of concurrent clients

100

Expected TPS

100

BACK

NEXT

ATTENTION: Playground environments are now deprecated. All existing playgrounds have been deleted. Please use PRE-PRODUCTION for any non-prod environment

IDPS Onboard

✓ The Basics — ✓ Application Details — ✓ Performance — 4 Extras

Additional users to be added to your IDPS project

Priti_Singh@intuit.com ketan_popat@intuit.com Mayank_Choubey@intuit.com Aditya_Bhardwaj@ir

Application administrator

Additional requirements

BACK

FINISH

ATTENTION: Playground environments are now deprecated. All existing playgrounds have been deleted. Please use PRE-PRODUCTION for any non-prod environment

- You will get the email/JIRA including your IDPS project endpoint.
- Create IDPS policy by
 - Click the IDPS project endpoint
 - Click API Access Policies
 - Click New Policy
 - Fill out the form like below for us-west-2 and us-east-2. You need to add IAM Role ARN used by Goldengate Hub

intuit

Data Protection

simon_liu@intuit.com

Manage Projects

User Profile

API Keys

API Access Policies

API Documentation

Downloads

About

Support

Logout

API Key Access Policies Management for SBG Production - Suhua Liu

API key policies allow applications to bootstrap their trust with the Intuit Data Protection Services (IDPS). The application obtains a restricted API key based on the application's characteristics

Policies

Show Active

New Policy

Create New Policy

Project *

QBORDS

Comment

Name or comment (free text)

Cloud Platform *

AWS

Region *

US West (Oregon)

Service

-- Not Selected --

Policy Rules ?

IAM Role ARN *

arn:aws:iam::094918931292:role/goldengate-iam-instance-profile-20-Ins

VPC ID

e.g. vpc-11726b73

Subnet ID

e.g. subnet-848fbcf0

Private IP Block

CIDR e.g. 192.168.0.0/1

Auto Scaling Group

Name of Auto Scaling gr

Maple Daemon restrictions

Linux Username

e.g. ec2-user

Linux Process Name

e.g. /bin/bash

Allowed Amazon Machine Image

Any Intuit baseline image

Only specified image

Only image (please provide reason)

Deny access if root device has been detached and re-attached ?

Save

Cancel

Search:

Created On

Created By Management Key

intuit

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API Key Access Policies Management for SBG Production - Suhua Liu

API key policies allow applications to bootstrap their trust with the Intuit Data Protection Services (IDPS). The application obtains a restricted API key based on the application's characteristics, and without needing to own an initial secret.

New policy created successfully with ID p-hpm3uwyyngxs

Policies

Show Active Policies

Show Expired Policies

New Policy

Edit Policy

Delete Policy

Search:

	ID	Project	Expires	Cloud	Comment	Owner	Created On	Created By Management Key
	p-hpm3uwyyngxs	QBORDS	never	AWS	Name or comment (free te	simon_liu@intuit.com	5/28/2019	

- You will get policy IDs. The following are the IPDS endpoints and policy IDs for PSP

Environment	Endpoint	us-west-2	us-east-2
Prod		p-xxxxx	p-xxxxx
PreProd		p-xxxxxx	p-xxxxx

- Enter the IPDS endpoints and policy IDs to the GG Hub config file like gg_hub_psp.config

```
["prod", "stg"]
```

```
ami_us_west_2="ami-0bc7d52924c776cc0"  
ami_us_east_2="ami-014d457eee8c38f79"  
vpc_env="prod"  
profile="sbg-psp-prod"  
goldengate_iam="arn:aws:iam::893547637742:instance-profile/goldengate-iam-instance-profile-20190805T115439-  
InstanceIamProfile-1w9CYJF6JJQU"  
notification_service_arn="arn:aws:sns:${region}:094918931292:gg-sbg-psp-prod-a-intuit-com"  
hosted_zone_id="Z1THM6G2H9YCW1"  
domain_name="sbg-psp-prod.a.intuit.com"  
idps_endpoint="vkm.ps.idps.a.intuit.com"  
idps_policy_id_us_west_2="p-xxxx"  
idps_policy_id_us_east_2="p-xxxx"
```

3. 3. Detail Steps

You should upgrade GG Hub on us-east-2 before us-west-2

3.1. 3.1 Upgrade GG Hub on us-east-2

- Download the following files https://github.intuit.com/payroll-services-platform/goldengate_hub/tree/master/scripts to your Mac. **Please always re-download the files for each db replication setup because they are keeping change for new feature or bug fix.**

```
create_gg_hub_ha.sh  
gg_hub_psp.config  
goldengate-hub-ha.json
```

- Remove auto start flag file on **us-east-2**

```
rm /u01/ogg/scripts/auto_start.flag
```

- Stop extract on **us-west-2**

```
GGSCI (ggpsppp001-a.sbg-psp-prod.a.intuit.com) 3> stop EPSP001
```

- Wait until no lag on pump on **us-west-2**

```
GGSCI (ggpsppp001-a.sbg-psp-prod.a.intuit.com) 2> send PPSP001 status
```

```
Sending STATUS request to EXTRACT PPSP001 ...
```

```
EXTRACT PPSP001 (PID 27206)
```

```
Current status: Recovery complete: At EOF
```

```
Current read position:
```

```
Sequence #: 5155
```

```
RBA: 288286941
```

```
Timestamp: 2020-02-18 22:22:20.000000
```

```
Extract Trail: /u01/ogg/psppp001/12.3.0.1.4/dirdat/psppp001/tr
```

```
Current write position:
```

```
Sequence #: 5158
```

```
RBA: 288391143
```

```
Timestamp: 2020-02-18 22:22:28.258733
```

```
Extract Trail: ./dirdat/psppp001/tr
```

- Stop pump on **us-west-2**

```
GGSCI (ggpspppib1.sbg-psp-prod.a.intuit.com) 7> stop PIBSP001
```

- Wait until all trail files are processed on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-psp-prod.a.intuit.com) 2> send RPSP001 status
```

```
Sending STATUS request to REPLICAT RPSP001 ...
```

```
Current status: At EOF
```

```
Sequence #: 5,158
```

```
RBA: 294,377,500
```

```
0 records in current transaction.
```

- Stop all processes on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-psp-prod.a.intuit.com) 8> stop *
```

- Stop manager on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-psp-prod.a.intuit.com) 9> stop mgr
```

- Run the following commands to save the credentials to /u01/tmp temporally on **us-east-2**

```
. /l/orcl
mkdir -p /u01/tmp
cp $GG_HOME/dircred/cwallet.sso /u01/tmp/dircred_cwallet.sso
cp $GG_HOME/dirwlt/cwallet.sso /u01/tmp/dirwlt_cwallet.sso
```

- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-b and goldengate-<RDS>-b in **us-east-2**. **Make sure you choose us-east-2 region.**
- Create new GG Hub with RHEL7 on **us-east-2**

```

chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
    where <AppName> should be psp, payments, etc
          <AppEnv> should be qa, e2e, stg, sbx, prf or prod
          <Region> should be us-west-2, us-east-2, etc
          <HA> should be yes or no
          <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
          [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

```

For prod, the command should like the below,

```

./create_gg_hub_ha.sh psp prod pspsp080 us-east-2 Intuit-vpc-3 yes
Note: This is for regular cluster db psppp080's standby db pspsp080.

```

or

```

./create_gg_hub_ha.sh psp prod pspsp099 us-east-2 Intuit-vpc-3 yes
Note: This is for small/testing cluster db psppp099's standby db pspsp099.

```

For non-prod, the following command creates E2E C3 GG Hub in us-east-2

- Restore the credentials and upload them to IPDS on GG Hub on **us-east-2**

```

. /l/orcl
APP_ENV=`cat /etc/intu_metadata/app.ini |grep APP_ENV |cut -d=' ' -f2`
ACCOUNT_ID=`cat /etc/intu_metadata/app.ini |grep ACCOUNT_ID |cut -d=' ' -f2`
IDPS_ENDPOINT=`cat /etc/intu_metadata/app.ini |grep IDPS_ENDPOINT |cut -d=' ' -f2`
IDPS_POLICY_ID=`cat /etc/intu_metadata/app.ini |grep IDPS_POLICY_ID|cut -d=' ' -f2`
AWS_REGION=`cat /etc/intu_metadata/app.ini |grep AWS_REGION |cut -d=' ' -f2`
CLUSTER_NAME=""`echo $RDS_SID`

cp /u01/tmp/dircrd_cwallet.sso /dev/shm/goldengate/${CLUSTER_NAME}/dircrd/cwallet.sso
cp /u01/tmp/dirwlt_cwallet.sso /dev/shm/goldengate/${CLUSTER_NAME}/dirwlt/cwallet.sso
cp /u01/tmp/password_file /dev/shm/goldengate/db/

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${CLUSTER_NAME}/dircrd/cwallet.sso --secret-value /dev/shm
/goldengate/${CLUSTER_NAME}/dircrd/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${CLUSTER_NAME}/dirwlt/cwallet.sso --secret-value /dev/shm
/goldengate/${CLUSTER_NAME}/dirwlt/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/db/password_file --secret-value /dev/shm/goldengate/db/password_file

```

- Start manager on **us-east-2**

```

GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 9> start mgr

```

- start extract and pump on **us-west-2**

```

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 14> start EIBPP001

```

```

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 15> start PIBSP001

```

- check and start replicat on **us-east-2**

```
GGSCI (ggpspspib1-a.sbg-ppsp-prod.a.intuit.com) 4> info RIBPP001
GGSCI (ggpspspib1-a.sbg-ppsp-prod.a.intuit.com) 3> start RIBPP001
```

- Setup auto start flag file on **us-east-2**

```
touch /u01/ogg/scripts/auto_start.flag
```

- Restore cronjob on **us-east-2**

```
crontab /u01/ogg/scripts/crontab.txt
```

- Delete the credentials on GG Hub on **us-aest-2**

```
rm /u01/tmp/dircrd_cwallet.sso
rm /u01/tmp/dirwlt_cwallet.sso
```

3.2. Upgrade GG Hub on us-west-2

- Remove auto start flag file on **us-west-2**

```
rm /u01/ogg/scripts/auto_start.flag
```

- Stop all extracts on **us-east-2**

```
GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 2> stop e*
```

- Wait until no lag on all pumps on **us-east-2**

```
GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 2> send PPSSP001 status

Sending STATUS request to EXTRACT PPSSP001 ...
EXTRACT PPSSP001 (PID 27206)
  Current status: Recovery complete: At EOF

  Current read position:
  Sequence #: 5155
  RBA: 288286941
  Timestamp: 2020-02-18 22:22:20.000000
  Extract Trail: /u01/ogg/psppp001/12.3.0.1.4/dirdat/psppp001/tr

  Current write position:
  Sequence #: 5158
  RBA: 288391143
  Timestamp: 2020-02-18 22:22:28.258733
  Extract Trail: ./dirdat/psppp001/tr
```

- Stop all pumps on us-east-2

```
GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 10> stop P*
```

- Wait until all trail files are processed on **us-west-2**

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 2> send RIBSP001 status
```

```
Sending STATUS request to REPLICAT RIBSP001 ...  
Current status: At EOF  
Sequence #: 7  
RBA: 4,824,877  
0 records in current transaction.
```

- Stop all processes on **us-west-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 8> stop *
```

- Stop manager on **us-west-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 9> stop mgr
```

- Run the following commands to save the credentials to /u01/tmp temporally on **us-west-2**

```
. /l/orcl  
mkdir -p /u01/tmp  
cp $GG_HOME/dirord/cwallet.sso /u01/tmp/dirord_cwallet.sso  
cp $GG_HOME/dirwlt/cwallet.sso /u01/tmp/dirwlt_cwallet.sso
```

- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-a and goldengate-<RDS>-b in **us-west-2. Make sure you choose us-west-2 region.**

- Create new GG Hub with RHEL 7 AMI on **us-west-2**


```

chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
    where <AppName> should be psp, payments, etc
          <AppEnv> should be qa, e2e, stg, sbx, prf or prod
          <Region> should be us-west-2, us-east-2, etc
          <HA> should be yes or no
          <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
          [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

```

For prod, the command should like the below,

```

./create_gg_hub_ha.sh psp prod psppp080 us-west-2 Intuit-vpc-3 yes
Note: This is for regular cluster db psppp080's standby db pspsp080.

```

or

```

./create_gg_hub_ha.sh psp prod psppp099 us-west-2 Intuit-vpc-3 yes
Note: This is for small/testing cluster db psppp099's standby db pspsp099.

```

For non-prod, the following command creates E2E C3 GG Hub in us-east-2

```

./create_gg_hub_ha.sh psp e2e epsppc03 us-west-2 Intuit-vpc-2 yes

```

Note: no volume size is needed since it will reuse the original storage

- Restore the credentials and upload them to IPDS on GG Hub on **us-west-2**

```

. /l/orcl
APP_ENV=`cat /etc/intu_metadata/app.ini |grep APP_ENV |cut -d=' ' -f2`
ACCOUNT_ID=`cat /etc/intu_metadata/app.ini |grep ACCOUNT_ID |cut -d=' ' -f2`
IDPS_ENDPOINT=`cat /etc/intu_metadata/app.ini |grep IDPS_ENDPOINT |cut -d=' ' -f2`
IDPS_POLICY_ID=`cat /etc/intu_metadata/app.ini |grep IDPS_POLICY_ID|cut -d=' ' -f2`
AWS_REGION=`cat /etc/intu_metadata/app.ini |grep AWS_REGION |cut -d=' ' -f2`
CLUSTER_NAME=`echo $RDS_SID`

cp /u01/tmp/dircred_cwallet.sso /dev/shm/goldengate/${CLUSTER_NAME}/dircred/cwallet.sso
cp /u01/tmp/dirwlt_cwallet.sso /dev/shm/goldengate/${CLUSTER_NAME}/dirwlt/cwallet.sso
cp /u01/tmp/password_file /dev/shm/goldengate/db/

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${CLUSTER_NAME}/dircred/cwallet.sso --secret-value /dev/shm
/goldengate/${CLUSTER_NAME}/dircred/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/${CLUSTER_NAME}/dirwlt/cwallet.sso --secret-value /dev/shm
/goldengate/${CLUSTER_NAME}/dirwlt/cwallet.sso

stash put-secret --api-key-grant-policy-id ${IDPS_POLICY_ID} --api-endpoint ${IDPS_ENDPOINT} --secret-name
secrets/goldengate/${APP_ENV}/${AWS_REGION}/db/password_file --secret-value /dev/shm/goldengate/db/password_file

```

- Start manager, extracts and pumps on **us-west-2**

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 4> start mgr
Manager started.
```

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 14> start EIBPP001

Sending START request to MANAGER ...
EXTRACT EIBPP001 starting
```

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 15> start PIBSP001

Sending START request to MANAGER ...
EXTRACT PIBSP001 starting
```

- check and start replicat on **us-west-2**

```
GGSCI (ggpspspib1-a.sbg-ppsp-prod.a.intuit.com) 4> info RIBSP001
```

```
GGSCI (ggpspspib1-a.sbg-ppsp-prod.a.intuit.com) 3> start RIBSP001
```

- Confirm the trail file is shipped from us-east-2 on us-west-2 and vice-versa
- Check and make sure GG replication is working from us-east-2 on us-west-2 and vice-versa
- Setup auto start flag file on **us-west-2**

```
touch /u01/ogg/scripts/auto_start.flag
```

1. Restore cronjob on **us-west-2**

```
crontab /u01/ogg/scripts/crontab.txt
```

- Delete the credentials on GG Hub on **us-west-2**

```
rm /u01/tmp/dircrd_cwallet.sso
rm /u01/tmp/dirwlt_cwallet.sso
```

4. Rollback

4.1. Downgrade GG Hub on us-east-2

- Remove auto start flag file on **us-east-2**

```
rm /u01/ogg/scripts/auto_start.flag
```

- Stop all extracts on us-west-2

```
GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 2> stop e*
```

- Wait until no lag on all pumps on **us-west-2**

```
GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 2> send PPSP001 status
```

```
Sending STATUS request to EXTRACT PPSP001 ...
```

```
EXTRACT PPSP001 (PID 27206)
```

```
Current status: Recovery complete: At EOF
```

```
Current read position:
```

```
Sequence #: 5155
```

```
RBA: 288286941
```

```
Timestamp: 2020-02-18 22:22:20.000000
```

```
Extract Trail: /u01/ogg/psppp001/12.3.0.1.4/dirdat/psppp001/tr
```

```
Current write position:
```

```
Sequence #: 5158
```

```
RBA: 288391143
```

```
Timestamp: 2020-02-18 22:22:28.258733
```

```
Extract Trail: ./dirdat/psppp001/tr
```

- Stop all pumps on **us-west-2**

```
GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 10> stop P*
```

- Wait until all trail files are processed on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 2> send RPSP001 status
```

```
Sending STATUS request to REPLICAT RPSP001 ...
```

```
Current status: At EOF
```

```
Sequence #: 5,158
```

```
RBA: 294,377,500
```

```
0 records in current transaction.
```

- Stop all processes on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 8> stop *
```

- Stop manager on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 9> stop mgr
```

- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-b and goldengate-<RDS>-b in **us-east-2**. **Make sure you choose us-east-2 region.**
- Create new GG Hub with RHEL 7 AMI on **us-east-2**

```

chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
    where <AppName> should be psp, payments, etc
          <AppEnv> should be qa, e2e, stg, sbx, prf or prod
          <Region> should be us-west-2, us-east-2, etc
          <HA> should be yes or no
          <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
          [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

```

For prod, the command should like the below,

```
./create_gg_hub_ha.sh psp prod psppp080 us-west-2 Intuit-vpc-3 yes
Note: This is for regular cluster db psppp080's standby db pspsp080.
```

or

```
./create_gg_hub_ha.sh psp prod psppp099 us-west-2 Intuit-vpc-3 yes
Note: This is for small/testing cluster db psppp099's standby db pspsp099.
```

For non-prod, the following command creates E2E C3 GG Hub in us-east-2

```
./create_gg_hub_ha.sh psp e2e epsppc03 us-west-2 Intuit-vpc-2 yes
```

Note: no volume size is needed since it will reuse the original storage

- Start manager on **us-east-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 9> start mgr
```

- start extract and pump on **us-west-2**

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 6> start EIBPP001
```

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 7> start PIBSP001
```

- start replicat on **us-east-2**

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 7> start RIBSP001
```

- Setup auto start flag file on **us-east-2**

```
touch /u01/ogg/scripts/auto_start.flag
```

- Restore cronjob on **us-east-2**

```
crontab /u01/ogg/scripts/crontab.txt
```

4.2. Downgrade GG Hub on us-west-2

- Remove auto start flag file on **us-west-2**

```
rm /u01/ogg/scripts/auto_start.flag
```

- Stop all extracts on **us-east-2**

```
GGSCI (ggpqbbogggpsppib1.sbg-psp-prod.a.intuit.compc04-a.sbg-qbo-ppd.a.intuit.com) 2> stop e*
```

- Wait until all trail files have been applied on **us-east-2**

```
GGSCI (ggpsppib1.sbg-psp-prod.a.intuit.com) 20> send PIBSP001 status
```

```
GGSCI (ggpqbbopc04-a.sbg-qbo-ppd.a.intuit.com as ggs@PQB0PC04) 24> stop PPQBSC04
```

```
Sending STOP request to EXTRACT PPQBSC04 ...  
Request processed.
```

```
GGSCI (ggpqbbopc04-a.sbg-qbo-ppd.a.intuit.com) 8> start mgr  
Manager started.
```

```
GGSCI (ggpqbbopc04-a.sbg-qbo-ppd.a.intuit.com) 9> start e*
```

```
Sending START request to MANAGER ...  
EXTRACT EPIAPC04 starting
```

```
Sending START request to MANAGER ...  
EXTRACT EPQBPC04 starting
```

```
GGSCI (ggpqbbopc04-a.sbg-qbo-ppd.a.intuit.com) 10> start p*
```

```
Sending START request to MANAGER ...  
EXTRACT PPQBSC04 startingStop pump on us-east-2
```

- Wait until all trail files are processed on **us-west-2**

```
GGSCI (ggpspsp001-a.sbg-psp-prod.a.intuit.com) 2> send RPSP001 status
```

```
Sending STATUS request to REPLICAT RPSP001 ...  
Current status: At EOF  
Sequence #: 5,158  
RBA: 294,377,500  
0 records in current transaction.
```

- Stop all processes on **us-west-2**

```
GGSCI (ggpspsp001-a.sbg-psp-prod.a.intuit.com) 8> stop *
```

- Stop manager on **us-west-2**

```
GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 9> stop mgr
```

- Use AWS Console to delete GG Hub Cloudformation stacks starting with goldengate-<RDS>-b and goldengate-<RDS>-b in **us-west-2**. **Make sure you choose us-west-2 region.**
- Create new GG Hub with RHEL 7 AMI on **us-west-2**

```
chmod +x create_gg_hub_ha.sh
./create_gg_hub_ha.sh
Usage: ./create_gg_hub_ha.sh <AppName> <AppEnv> <DBName> <Region> <VPC> <HA> [<VolumeSize>]
      where <AppName> should be psp, payments, etc
            <AppEnv> should be qa, e2e, stg, sbx, prf or prod
            <Region> should be us-west-2, us-east-2, etc
            <HA> should be yes or no
            <VPC> should be vpc-1, vpc-2, etc or Intuit-vpc-1, Intuit-vpc-2, etc
            [<VolumeSize>] is optional. It is EBS volume /u01 size in GB and default is 1000 GB

For prod, the command should like the below,

./create_gg_hub_ha.sh psp prod ppspsp080 us-east-2 Intuit-vpc-3 yes
Note: This is for regular cluster db psp080's standby db ppspsp080.

or

./create_gg_hub_ha.sh psp prod ppspsp099 us-east-2 Intuit-vpc-3 yes
Note: This is for small/testing cluster db psp099's standby db ppspsp099.

For non-prod, the following command creates E2E C3 GG Hub in us-east-2
```

- Start manager, extracts and pumps on **us-west-2**

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 8> start mgr
Manager started.
```

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 9> start e*
```

```
Sending START request to MANAGER ...
EXTRACT EPIAPC04 starting
```

```
Sending START request to MANAGER ...
EXTRACT EPQBPC04 starting
```

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 10> start p*
```

```
Sending START request to MANAGER ...
EXTRACT PPQBSC04 starting
```

- [start the replicat](#) on **us-west-2**

```
GGSCI (ggpspppib1.sbg-ppsp-prod.a.intuit.com) 11> start PIBSP001
```